LUIS OLEA

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TECHNICAL SKILLS

Languages: C/C++ (2+ yrs), Python (2+ yrs), JavaScript/TypeScript (9 mo), Java (3 mo)

Tools & Frameworks: React, Next.js, Supabase, Git, Figma, Linux, Pandas, NumPy, Matplotlib, PyTorch, Scikit-learn

Relevant Coursework: Algorithms & Data Structures, Computer Organization, Object-Oriented Programming, Discrete Mathematics, Calculus I–III, Linear Algebra

PROJECTS

Ball Up Top — Founder & Developer | ***Next.js, React, Supabase*** **July 2026 – Present**

- Founded and built a Letterboxd-inspired social app to track and rate games across the NBA, NFL, MLB, and UFC; deploying to Android in early November.
- Designed the Supabase schema and backend for user profiles, ratings, and game data across major sports leagues; built the Next.js/React frontend end-to-end as sole engineer.
- Used a public API (balldontlie.api) to backfill previous-match history.

Future Car Depreciation Calculator — MLSN (Machine Learning Club, UC Davis) | ***Python, Pandas, CatBoost, Streamlit*** **January 2026**

- Built a CatBoost regression model in Python predicting vehicle depreciation from mileage, condition, model year, and historical pricing data.
- Engineered frequency encoding and multi-year aggregation features; cleaned 250,000+ vehicle records to reduce model error by around 40%.
- Shipped the model behind a Streamlit frontend.

Open-Source Contribution Tracker — CodeLab, UC Davis | ***React, TypeScript, Figma*** **June 2026**

- Built the frontend for a leaderboard ranking contributions across CodeLab's open-source projects; used to recruit new members.
- Implemented Figma designs in React/TypeScript on a 9-person team; owned the ranking algorithm and individual profile implementation.

Tetris Neural Network — Deep Reinforcement Learning | ***Python, NumPy*** **August 2026**

- Built a neural network from scratch in NumPy, trained via deep reinforcement learning to select optimal Tetris moves from raw board state.
- Implemented a reward function and backpropagation pipeline; after over 2000 training episodes, the agent achieved a 99% improvement over a random-move baseline.

EXPERIENCE

Computer Programming Teacher | ***Breakthrough Sacramento*** **June 2026 – August 2026**

- Designed and taught an Intro to Python curriculum (variables, conditionals, loops, lists, final calculator project) to 8th/9th-grade students.
- 100% of students completed the final project; around 80% left more passionate/curious about programming.

Sports Supervisor | ***UC Davis*** **July 2025 – Present**

- Supervise recreational and club sporting events for participants and viewers; first point of contact for injuries and conflicts.
- Coordinate communication between officials, participants, and staff to resolve disputes and enforce safety protocols and institutional policy.

EDUCATION

B.S in Computer Science at University of California, Davis **September 2024 – Present**